

3

intelligent machines

Mark Iyobak-

algorithm - follow steps

father of computation → Turing → Math guy → universal computing machine
1954

gave right instruction
that computing machine
would substitute any
other -

Before the function was producing intelligent
behaviour

Now: finding the right algorithm

before - symbols + rules → like language

now

distributed patterns of activity in a network inspired by
the human brain → connectionist networks -

→ Maybe computation
can unravel

human's intelligence

how the brain works

probabilistic networks

allow $\leftarrow$ array of outcomes
uncertainty of those outcomes
system to learn from experience

Bayes Rule

1970s - Marr → which task does the brain solve

how does the brain solve that task

why is that task important

computational level

algorithm -

coding level

1° Computational level → which
why

2° Algorithmic → how

we try to find out:
how which steps it takes

imagine it

how fast, how much memory it has

we test hypothesis
↓
we form a hypothesis
↓
test it

3° Implementation level → how the physical parts interact to achieve the algorithm? which is ~~the~~ the role of every part?

intervene in steps

observe changes when executing algorithm -

mapping changes to steps on algorithm

Cognitive Scientist follow a similar logic for behaviour
which computation does the brain perform
which algorithm uses
which physical parts are relevant for its implementation

fit $\rightarrow$ hypothesis about
algorithm that
brain is
running

$\rightarrow$ data about human reaction times
and susceptibility to error

3.2

Peggy Series

$\rightarrow$ Brain as a probabilistic machine

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Hermann von Helmholtz

$\downarrow$ was studying the eye $\rightarrow$ imperfect visual instrument

visual perception $\rightarrow$ guess of unconscious inference

$\downarrow$

the brain completes missing information using past
knowledge

$\downarrow$

Constructs an hypothesis about our environment

"reality"

$\downarrow$

Conditional probabilities

"p"

|

we can get
the probability
P

$\leftarrow$

Bayes
Rule

$\leftarrow$

the probability of the hypothesis
given the data

$\downarrow$
 $P(H | D)$

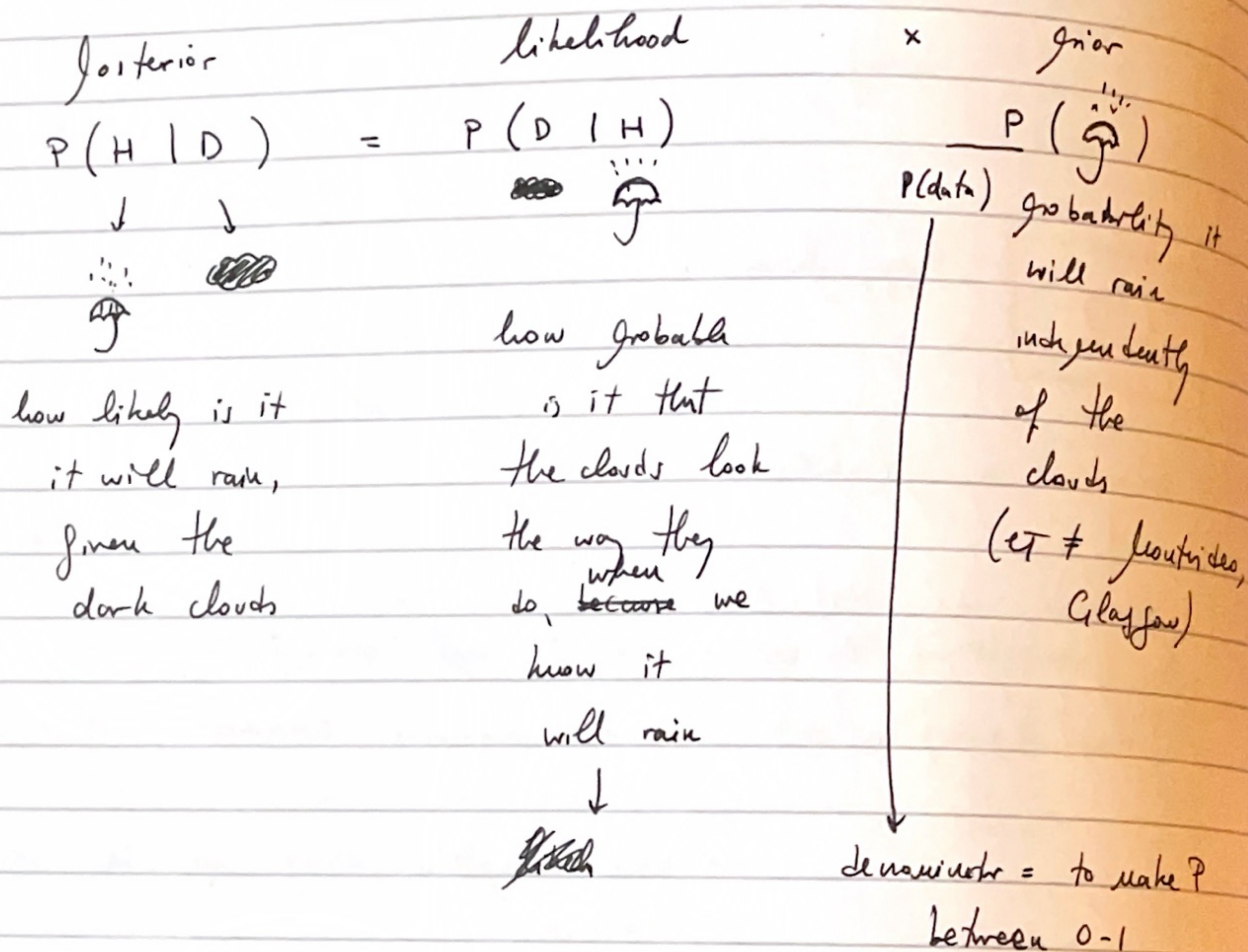
signals available to
our senses

by multiplying other two
probabilities $\rightarrow$

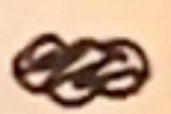
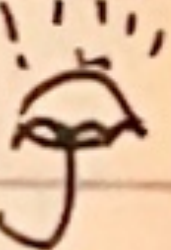
$\downarrow$

()

likelihood $\rightarrow$ prior probability $\rightarrow$ or prior knowledge before we collect any new info



$$P(H | D) = \frac{P(D | H) P(H)}{P(D)}$$

D = 
H = 

$\rightarrow$ The prior and the likelihood form an internal model of the ground, a model of the world inside our brain

we use to interpret the extended environment

$\rightarrow$ how do we know if our brain uses Bayesian inference

McGurk effect $\leftarrow$ there can lead to illusions

$\leftarrow$ predicts the optimal is to combine information from $\neq$ senses
weight it according to reliability

→ Predictions from Bayesian model and human performance

show our brain works similarly to Bayesian brains

predicts → how to combine & process of info
→ how to incorporate prior knowledge → in terms of

particularly when uncertainty is ↑

if the brain uses prior beliefs

→ which are those

→ how do they influence perception

prior distribution
(summary of all previous experience)

should be
multiplied with
incoming information

→ the reality that we perceive is consistent with our expectations but not with the physical world

→ our brain constantly updates its understanding of the world → its "model"

Current research -

→ how these ~~are~~ algorithms are implemented in the brain

→ can Bayesian models make predictions in neurobiology?